

Ex:- 123456789103(5)  $\rightarrow$  Is it divisible by 5?

Ans:- Yes

$\because$  because last digit is 5 & 0.

123456789103(0)  $\rightarrow$

Ex:- 1234567891035  $\rightarrow$  Is it divisible by 3?

Sol<sup>n</sup> - first add the numbers.

$$1+2+3+4+5+6+7+8+9+10+3+5 = 48 = 4+8 = 12$$

$$\text{Ans} = 12$$

then  $\div 3$

$$= \frac{12}{3} \Rightarrow \text{there for it is } \div 3$$

for divisibility

$$0 \left\{ \begin{array}{l} \frac{0}{1}, \frac{0}{2}, = 0 \end{array} \right.$$

$$\frac{1}{0}, \frac{2}{0}, = \text{Not defined.}$$

$$1 \rightarrow \text{unique } \frac{1}{1}, \frac{2}{1}, \frac{1}{2}$$

No problem for  $\div 6$

the last digit is '0'

$$2 \left\{ \begin{array}{l} 0 \\ 2, 4, 6, 8 \end{array} \right.$$

all even no are  $\div$  by 2

$$4 = (2^2) \rightarrow \text{last two digits } \div 4$$

Ex:- 25144  $\div 4$ ?

$$\text{Sol}^n: \frac{44}{4} = 11$$

yes it is  $\div$  by 4



$8 \Rightarrow (2^3) \rightarrow$  last '3' digit is  $\div$  by 8 it is  $\div$  by 8 (Whole no / all no)

$16 \Rightarrow (2^4) \rightarrow$  last '4' digit is  $\div$  by 16 it is also  $\div$  by 16 (Whole no / all no)

3  $\rightarrow$  Sum of digits = Ans / 3

9  $\rightarrow$  Sum of digits = Ans / 9

num  $\div 9$  is also divisible by 3

but reverse is not true

num  $\div 3$  may be  $\div$  by 9 but not all

6  $\rightarrow$  1 2 3

$\downarrow$   
even

$\downarrow$

In the 3's table

7  $\rightarrow$  1 2 (7)

$7 \times 2 = (14)$

double it

$12 - 14 = -2$

$\Rightarrow$  not  $\div$  by 7

Sn:-

14 (7)

$7 \times 2 = 14$

$14 - 14 = 0$

$\therefore$  it is  $\div$  by 7.

Note:-

Last digit  $\times 2 =$  Ans

Remaining part no. - Ans = 0 / 7 / 7 multiple



10  $\rightarrow$  last digit '0'

5  $\rightarrow$  last digit is 0 or 5

13  $\rightarrow$  last digit  $\times \underline{4} = \text{Ans}$

RPN + Ans  $\equiv 0/13/13$  multiples  
(Remaining part no.)

13  $\rightarrow 3 \times 4 + 1 = 13$   
 $3x + 1$

19  $\rightarrow$  Last digit  $\times 2 = \text{Ans}$

RPN + Ans  $\equiv 0/19/19$  multiples

29  $\rightarrow$  LD  $\times 3 = \text{Ans}$

RPN + Ans  $\equiv 0/29/29$  multiples

11  $\rightarrow$

• 1221

$$1 - 2 = -1$$

$$2 - 1 = +1$$

0/11/11 multiples

an

$\begin{array}{cccc} 1 & 2 & 2 & 1 \\ \hline 1 & 2 & 3 & 4 \\ \text{odd} & \text{even} & \text{odd} & \text{even} \end{array}$

$\rightarrow$  places.

(Sum of odd place) - (sum of even place)  $\equiv 0/11/11$  multiples

$$= (1 + 2) - (2 + 1)$$

$$= (3) - (3) = 0 \quad \checkmark$$

$$\therefore \frac{2}{11}$$



Sx:- 121

$= (1+1) - (2)$

$= 2 - 2 = 0 \therefore 11 \div 11$

25  $\rightarrow$  last two digit  $\div 25$

125  $\rightarrow$  last three digit  $\div 125$

### Question 5

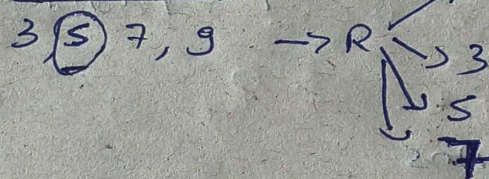
(asking for which no. is gives remainder)

① The greatest number of four digits which when divided by 3, 5, 7, 9 leave remainders 1, 3, 5, 7 respectively is: (Divisibility rule of 3, 5, 7 & 9)

Sol<sup>n</sup>

taken

~~3763~~



a) 9763

b) 9766

c) 9764  $= 26 \div 8$

$\begin{array}{r} 376 \\ -18 \\ \hline 968 \end{array}$

x d) 9765  $= 0 (\div 5)$

for 5

the above / greater than 5 no. is  $\div 5$  & gives Remainder

that's are

$\frac{6}{5} = R \rightarrow 1$

$\frac{7}{5} = R \rightarrow 2$

$\frac{8}{5} = 3$

$\frac{9}{5} = 1 \frac{4}{5}$   
 $\frac{10}{5} = 2$

$\frac{13}{5} = 2 \frac{3}{5} \rightarrow 3$

$\therefore$  the last digit is 3 in the option that's way the ~~R~~ must be give 3

for another method

$\overset{+}{9}\overset{+}{7}\overset{+}{6}3 \rightarrow 25 \div 3 \rightarrow R \rightarrow 1$

$25 \div 9 \rightarrow R \rightarrow 7$

$\therefore$  Ans is (a)

$\therefore$  Ans :- a) 9763



- ② 8 years ago shreya was equal to the sum of the present ages of her one son and one daughter. Five years hence the respective ratio b/w the age of her daughter & her son that time will be  $7:6$ . If Shreya's husband is 7 years older to her & his present age is three times the present age of their son, what is the present age of the daughter? ( $\div$  Rule of 7)

Sol<sup>n</sup>:

5 yrs later

$7:6$   
Daughter Son

$7:6$

$\therefore \div 7$

NOTE

for hence +

for ago -

a) 15 years  $+5 = 20$

b) 23 y  $+5 = 28$  ✓  $28 \div 7 = 4$

c) 19 y  $+5 = 24$

d) 13 y  $+5 = 18$

Ans is (b) 23 years

- ③ Total no. of stds in a school was 660 Ratio b/w boys & girls was  $13:9$ . After some days, 30 girls joined the school & some boys left the school. Now the new ratio b/w boys & girls is  $6:5$ . Find the no. of boys who left the school?

a) 50

b) 30

c) 40

d) 60

Sol<sup>n</sup> Given

$660 - 50 = 610$

$660 - 30 = 630$

$660 - 40 = 620$

$660 - 60 = 600$

W.K.T  $\div$  11  $\Rightarrow$  0/11/11 multiple

$660 \rightarrow$  1st all stds  $\rightarrow 13:9$

$30 \rightarrow$  add girls

$660 - x$

$\rightarrow 6:5 = 11$

some boys are left

Put the x value

as options & the

check the ans is  $3 \div 11$

③

③



W.K.T

11  $\rightarrow$  (sum of odd digits) - (sum of even digits)

$$\frac{640}{1\ 2\ 3} = (6+0) - (4) = 2 \quad \times$$

$$660 = (6+0) - (6) = 0 \quad \checkmark$$

$\therefore$  Ans (B) 30

④ The ratio of number of boys to the no. of girls in a school of 432 pupils is 5:4. When some new boys and girls are admitted, the no. of boys increase by 12 & the ratio of the boys to the girls changes 7:6. Find the no. of new girls admitted.

a) 14

b) 24

c) 12

d) 20

Given

B:G

5:4  $\rightarrow$  432

B  $\rightarrow$  +12

N.T. 444 + x

$\frac{x}{5}$  no. girls add.

New ratio : 7:6

7+6 = 13

$\therefore$  444 + x

(then the ans  $\div$  13 that is the Ans)

$$a) 444 + 14 = 458$$

$$b) 444 + 24 = 468$$

$$c) 444 + 12 = 456$$

$$d) 444 + 20 = 464$$

$$458 \rightarrow 8 \times 4 = 32 \rightarrow 45 + 32 = 77 \quad \times$$

$$468 \rightarrow 8 \times 4 = 32 \rightarrow 46 + 32 = 78 \quad \checkmark$$

$$456 \rightarrow 6 \times 4 = 24 \rightarrow 45 + 24 = 69 \quad \times$$

$\therefore$  Ans (B) 24

W.K.T

13  $\div$  75

Last digit  $\times$  4 = Ans

Remaining part + Ans = 0/13/13 mult.



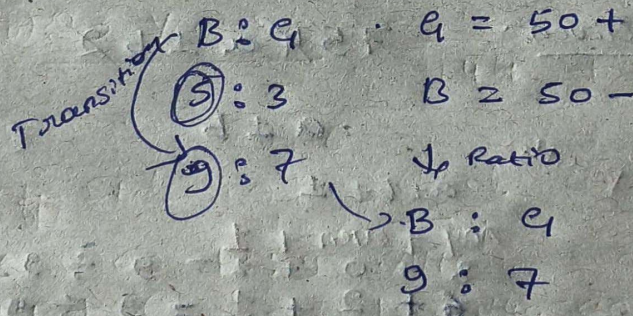
Q) Among 132 examinees of certain school, the ratio of successful & unsuccessful stds is 9:2. Had four stds passed then the ratio of successful & unsuccessful stds would have been?

- a) 14:3  $\rightarrow 17 \times$  Sol<sup>n</sup> Given :- 132 ex - -
- b) 14:5  $\rightarrow 19 \times$   $9:2 = 9+2 = 11$
- c) 28:5  $\rightarrow 33 \checkmark$
- d) 28:3

$\therefore$  Ans 28:5

Q) The ratio of boys & girls in a college is 5:3. 50 boys leave the college & 50 girls join the college. The ratio becomes 9:7. The no. of boys in the college is: (Rule 5 & 9)

- a) 600
- b) 300
- c) 400
- d) 500



In the question the 50 boy leave the coll.  
 $\therefore$  find the no. boys

- a)  $600 - 50 = 550$
- b)  $300 - 50 = 250$
- c)  $400 - 50 = 350$
- d)  $500 - 50 = 450 = 4+5+0 = 9 \div 9$

$\therefore$  Ans d) 500



Q) 71352  $\times$  608 is divisible by 3 then the maximum possible digit in the place of  $\times$  is : (Rule 3)

a) 9

b) 3

c) 5

d) 7

Given:-  $71352 \times 608 \Rightarrow 18 + x + 14 \Rightarrow 32 + x$   
 $\Rightarrow 5 + x$

w.k.T

$\div 3$  is adding all nos & that no. is  $\div 3$

a)  $32 + 9 = 5 + 9 = 14$

b)  $5 + 3 = 8$

c)  $5 + 5 = 10$

d)  $5 + 7 = 12 \rightarrow \div 3$

$\therefore$  Ans - (d) 7

Note:-

$(5 + x) \div 3$

1  
4  
7

Q) How many possible values of  $a$ , where be is an even digit in the number 754a1b2. If the number is divisible by 9. (9's sum  $\div 9$  rule)

a) 8

b) 9

c) 6

d) 5

Sol<sup>n</sup> Given:-  $754a1b2$

$a + b$

(by 9's Remainder method)

Possibles

$a + b + 1$

Any + even + 1

$8 + 0 + 1 = 9 \div 9$

$6 + 2 + 1 = 9 \div 9$

$4 + 4 + 1 = 9 \div 9$

$2 + 6 + 1 = 9 \div 9$

$9 + 8 + 1 = 9 \div 9$

(10 is 2 digit)

Ans = (d) 5



Q) Find the remainder when  $1011121314 \dots 9899$  is divisible by 8

a) 5

b) 4

c) 3

d) 2

Sol<sup>n</sup> take last 3 digits

$$\begin{array}{r} 8 \overline{) 899112} \\ \underline{8} \phantom{112} \\ 09 \phantom{112} \\ \underline{8} \phantom{112} \\ 19 \phantom{112} \\ \underline{16} \phantom{112} \\ 03 \end{array}$$

Ans c) 3

Q) What is the value of  $a$  such that  $51763a4$  is divisible by 12

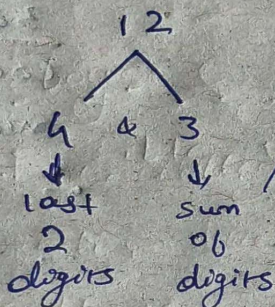
a) 7

b) 8

c) 4

d) 1

Sol<sup>n</sup>



Take sum of digits

$$51763a4 = 26 + a$$

$$\checkmark 26 + 7 = 33 \div 3$$

$$\times 26 + 8 = 34 \div 3$$

$$\checkmark 26 + 4 = 30 \div 3$$

$$\checkmark 26 + 1 = 27 \div 3$$

or,

$$26 + \boxed{4} = 30 \div 3$$

↓

for the 4 it will be  $\div 3$  &

also the no.  $51763\overset{a}{4}4$

$\rightarrow \div 4$

$$26 + a = 8 + a$$

$$\cancel{8 + 7}$$

Ans c) 4



Q) The 6 digit number  $439xy5$  is divisible by 125.  
How many such six digit no.'s are there?

- a) 5
- b) 2
- ~~c) 4~~
- d) 3

Given :-  $439xy5 \rightarrow \div 125$

They asking How many such digits.

odd mult

$$\begin{aligned} 125 \times 1 &= 125 \rightarrow \checkmark \\ 125 \times 2 &= 250 \times \\ 125 \times 3 &= 375 \rightarrow \checkmark \\ 125 \times 5 &= 625 \checkmark \\ 125 \times 7 &= 875 \checkmark \\ 125 \times 9 &= 1125 \end{aligned}$$

Here it is 4 digit  
but the xy5 are  
3 digits the 5 is  
common only x & y  
means 2 digit before 5.  
∴ here 3 digit before 5  
∴ It is not considered.

∴ Ans c) 4

Q) How many 5 digit numbers of the form  
 $xxxyxx$  are divisible by 44?

- a) 1
- b) 3
- ~~c) 2~~
- d) Infinite

Sol<sup>n</sup>

$$\begin{array}{cccccc} x & x & y & x & x & \\ 1 & 2 & 3 & 4 & 5 & \end{array} \div 44$$

W.K.T  $11 \div$  rule

$$\begin{aligned} &= (\text{sum of odd}) - (\text{sum of even}) \\ &= (x + y + x) - (x + x) \\ &= x + y + x - 2x \\ &= 2x + y - 2x \\ &= y = 0 \end{aligned}$$

$$\begin{array}{r} 44 \\ \wedge \\ 11 \times 4 \end{array}$$

or method.

$$\begin{array}{cccccc} x & x & y & x & x & \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \end{array}$$

$$x - x = 0$$

$$y - x = y - x$$

$$+ x =$$

$$0 + y - x + x =$$

$$\therefore y = 0.$$

$$\therefore xx0xx$$

$$44 \div 44 \quad (\because \div 4)$$

$$88 \div 88$$

for 99 0 99 it only  $\div 11$  but not  $\div 4$

∴ These two are  $\div$  by 11 & 4 both.

Ans = c) 2